

Design and Implementation of Student Attendance System Based on Face Recognition by Haar-Like Features Methods

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Abstract— The attendance system is very important for every organization because it can track the people involved in it to maximize their performance. There are various methods in applying the presence system, including using faces, fingerprints, iris, sounds, and so forth. The ability to manage student attendance during attendance is a challenge because errors often occur and take a long time. In the traditional system, it is usually done by signing a piece of paper that is carried by the lecturer in each class session. This research presents the development of student attendance systems using facial recognition patterns with the Haar-like features method. With this method, it can overcome several manual system problems and provide solutions such as limits on the delay for lecturers and students, as well as minimizing the possibility of fraud in the attendance system. In the Administration section with this model can be automated to save time and resources needed to enter attendance data. After the proposed model was implemented, it was proven to reduce the time needed to enter attendance data from 6 until 8 days to only 3 to 4 days.

Keywords— *Haar-like features, face recognition, student attendance system, raspberry pi*

I. INTRODUCTION

The attendance system is very important for every organization because it is useful to control the presence of each member. There are methods that can be applied. Some apply it manually, namely by signing attendance papers, which of course takes a lot of time and a waste of resources. Also there is often cheating in the absence process and has the risk of human error, and there is also an automatic attendance system. The automatic attendance system includes many methods such as Quick Response Code (QR-Code) technology[1], Radio Frequency Identification (RFID)[2], using fingerprints[3], using iris[4], as well as using face recognition[5].

From these studies, there are two methods used, namely biometrics and non-biometrics. Biometrics is a method for automatically recognizing one's identity based on physical and behavioral characteristics[6]. Meanwhile, QR-Code and RFID technology are based on non-biometrics. The basis for identifying QR-Code technology is a barcode that can be read using a scanner, while the basis for identifying RFID technology is radio waves that can be read using a wave reader. QR-Code and RFID technology has several advantages, namely its easy use and very high accuracy. However, QR-Code and RFID also have weaknesses, that is if students forget to bring object identification, then the attendance process cannot be carried out. Meanwhile, in

biometric technology, the basis of identification uses physical characteristics and human behavior.

Characteristics and physical behavior of humans are always attached to the human body unless it is lost or damaged due to aging, accidents, or because it was born defective. This makes biometric technology far more practical than non-biometric technology which requires the user to always carry object identification.

In this paper, an application for attendance based on biometric technology has been developed, namely by applying an attendance registration system that uses face identification as input data, and then analyzes the level of accuracy by the system. The system is built using a web camera (webcam) that is integrated with the Raspberry Pi as a control unit to make attendance. The aim of developing this system is to provide the easiest and most effective solution for recording and recording attendance. This system is predicted to be very cheap from systems that are already on the market, compared to prices on the market with almost the same quality, the price can be reduced by around 60%.

II. METHOD

During this time at the Universitas Prima Indonesia, especially in the information systems department, attendance systems are still carried out conventionally by signing attendance papers that are brought by lecturers in each lecture. It is certainly less effective because of the opening of opportunities to commit fraud among students themselves and there is no definite delay limit from the university and every lecturer who teaches. In addition, arrangement by the academic administration is also done in a conventional way that takes a long time just because it includes a lot of data[7].

To solve the problem, the attendance registration system can be applied using biometric technology in the form of a presence system that uses face identification as input and analyzes the accuracy of the recognition made by the system. By utilizing the pattern of physical characteristics, this thesis is investigated to recognize a student, namely by facial recognition. The system built utilizes a web camera (webcam) on a computer for attendance[8]. The aim of developing this system is to provide the easiest and most effective solution for recording and recording lecture attendance at Universitas Prima Indonesia[9].

In this paper, to recognize the model of the human face, the Bilateral Filter, Haar-like features[8] method will be used. To measure the effectiveness of the proposed method, the SFAM-NN method will be compared to the Cascade

Classifier Adaboost method for adaptive boosting to fix the classification problems.

A. Face Recognition

The face recognition algorithm of the human face consists of several steps namely feature extraction, training and database matching. Training is a process to record the face pattern to be recognized and labeling the face. Each label of the face should have multiple instances of face samples stored in the database[10]. The process of face recognition is shown in Fig 1.

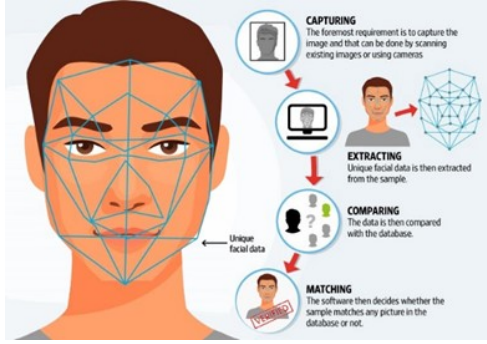


Fig. 1. Face Recognition Process

B. Haar-like features Methode

The Haar-like features method is a rectangular feature, which gives a specific indication to an image. This method has the advantage that the computation is very fast because it only depends on the number of pixels in a square not every pixel value of an image[11][12].

Image data training requires 2 types of object images in the training process, which are: Positive samples & Negative samples. This method is determined by subtracting the average pixel in dark areas from the average pixel in bright areas. The value of haar-like features is the difference between the gray level pixel values in the black box and white box regions, the gray level equation in the haar-like feature is as follows:

$$\hat{f}(\chi) = \text{SumBlack Rectangle} - \text{SumWhite Rectangle} \quad (1)$$

C. Cascade Classifier AdaBoost

AdaBoost is an adaptive boosting algorithm that aims to improve the accuracy of the given learning algorithm, a weak classifier with a high error rate, iterates to reduce the level of misinformation so that a strong classifier is formed. Cascade classifier results in increased detection by reducing computing time. Boosting classifier is created by removing negative images when detecting positive images. Classification is created in stages to get rid of negative images that are not needed in the process before detecting more complex objects to get the error rate of detecting positive images. The model of the cascade classifier AdaBoost are shown in Fig. 2

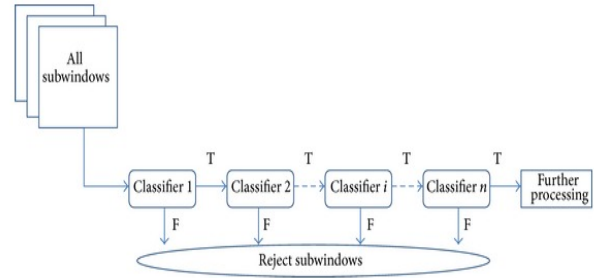


Fig. 2. Cascade Classifier Adaboost Models

D. Bilateral Filter

Bilateral Filter is a filtering technique that refines an image by maintaining the edge sharpness of the image. In general, the image has 3 color combinations namely: red, green, and blue, or commonly referred to as RGB. The smoothness process separately will disrupt the color balance, therefore the bilateral filter will process all three colors simultaneously and determine the middle value based on the color match. Calculation of range values based on two pixels similarity is determined based on the number of low pass filter values. Low pass domain filters can be applied to the following equations:

$$h(x) = k_d^{-1}(x) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\xi) c(\xi, x) d\xi \quad (2)$$

Where, $c(\xi, x)$ measures the geometric proximity between the center of x and the nearest point ξ . For f and h emphasize the fact that both input and output images may be multiband. $h(x)$ is the image output, and is the normalized constant value calculated as the following equation:

$$k_d(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c(\xi, x) d\xi \quad (3)$$

E. Canny Edge Detection

An edge is a change in the value of degrees of intensity that is fast or sudden within a short distance. The purpose of detecting edges is to group objects in the image, and also to analyze the image further. The Canny algorithm runs in 5 separate steps:

- Smoothing
Blur the image to eliminate noise

$$H_{ij} = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(i-(k+1))^2 + (j-(k+1))^2}{2\sigma^2}\right); 1 \leq i, j \leq (2k+1) \quad (4)$$

- Finding Gradient
Edges should be marked on images that have large gradients.

$$G = \sqrt{G_x^2 + G_y^2} \quad (5)$$

$$\theta = \text{atan2}(G_y, G_x) \quad (6)$$

- Non-maksimum-suppression
Only local maximums must be marked as edges.

$$\text{Edge_Gradient}(G) = \sqrt{G_x^2 + G_y^2} \quad (7)$$

$$Angle(\theta) = \tan^{-1} \left(\frac{G_x}{G_y} \right)$$

- Double thresholding
Edge with thresholding potential.
- Edge Tracking by hysteresis
The final edge is determined by pressing all the edges that are not connected with a very strong edge.

F. Software Development Life Cycle

Making attendance application system with facial recognition consists of several stages, namely identification of the problem of recording systems of students who are still

- (8) using conventional systems, studying the literature obtained from journals and books about the system of recording student attendance by face recognition, collecting facial pattern data as a dataset training from the subject, designing hardware using raspberry pi, camera, MicroSD, designing and building a student attendance recording system with facial recognition features using the website-based python programming language, conducting face recognition patterns for the subject using the haar-like feature method and the final stage is system simulation that has been designed. The stages of research methods are shown in Fig. 3.

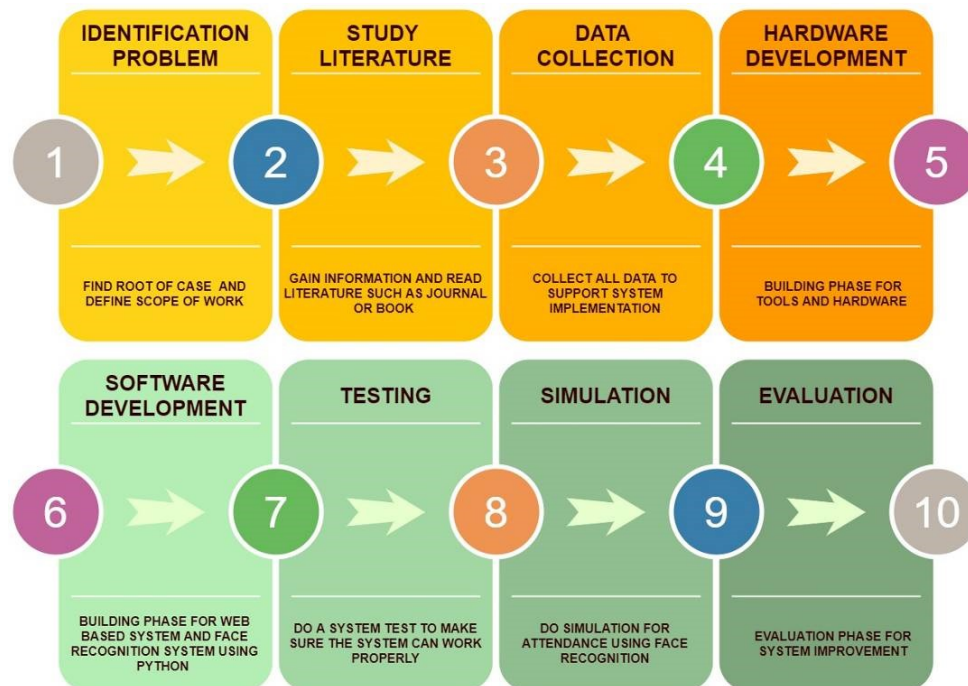


Fig. 3. Stages of Research Methods

In this study, the attendance application system with facial recognition consists of several supporting tools which are Microprocessor Raspberry Pi 3 B, Raspberry Pi Wide Angle Fisheye Lens Camera, Memory Micro SD, Monitor, Keyboard, Mouse, Power Adapter, HDMI cable, Converter HDMI to VGA.

The first stage of the process is the stage of preparing data, which is the process of taking data from students. Data taken from students can be in the form of names, student ID numbers, semesters, majors, and so on. Furthermore, the process of taking face patterns from each student. The process is needed so that during the implementation of the system, students can attend the attendance process because

the data from their face patterns are already connected to the database. The process requires training data where the facial pattern taking of each student will be done 30-50 times so that the accuracy of the resulting system is increased.

The next stage is the synchronization process of the face pattern recognition system into the student academic system. In this study, the system that was built still needed a manual checking process carried out by the academic admin to avoid mistakes in the synchronization process into the academic system. Flow diagrams of student attendance systems using face recognition are shown in Fig. 4.

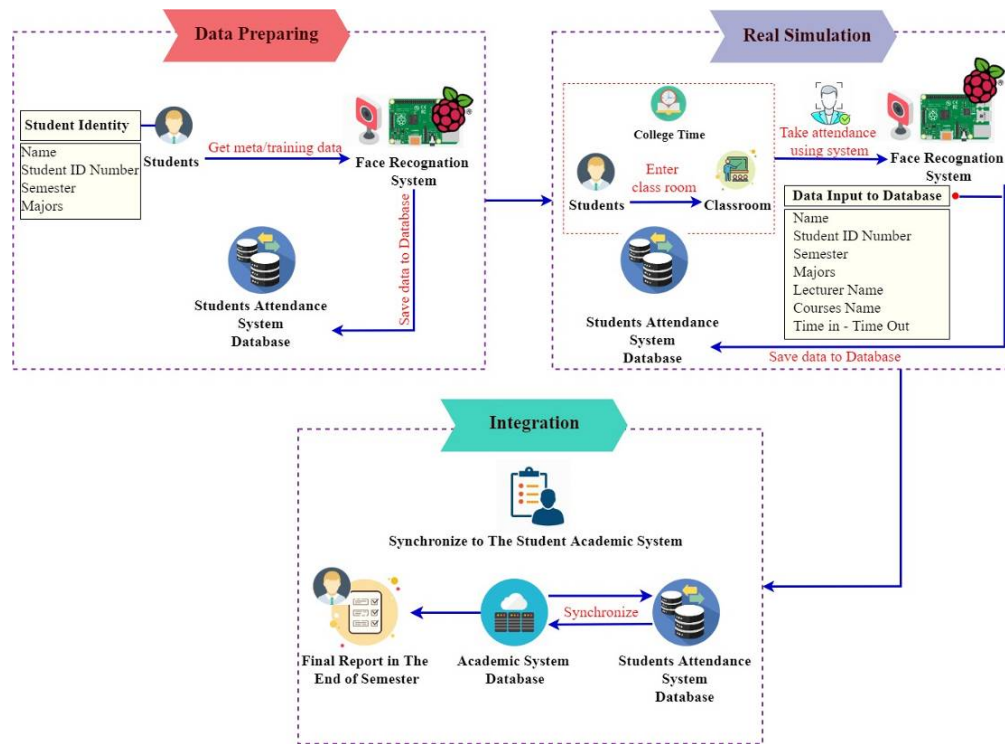


Fig. 4. Flow Diagram of Student Absention System Using Face Recognition

The use case of the system built involves two actors namely admin and student. Admin has the duty to carry out the process of preparing data from students as well as the recap stage of student attendance. Students can only attend the attendance process if the student's personal data has been verified. In Fig. 5, it can be seen that the admin can do several processes contained in the attendance system.

The first process is inputting student data, where the admin is in charge of preparing the data from students in the form of the student ID number, name, department, semester, and conducting the training stages of the student's facial patterns so that the data is stored in a database.

The next process is inputting attendance data. Because this system is connected to the database, the admin only has the duty to view and cross-check daily attendance. The third process is the synchronization process of attendance of the system designed into the academic system of students.

In this case, the same admin will do the process by manually inputting through the academic system. And the last process is making a weekly report that attendance is completed in a week.

Use case diagrams of student absence systems using face recognition are shown in Fig. 5.

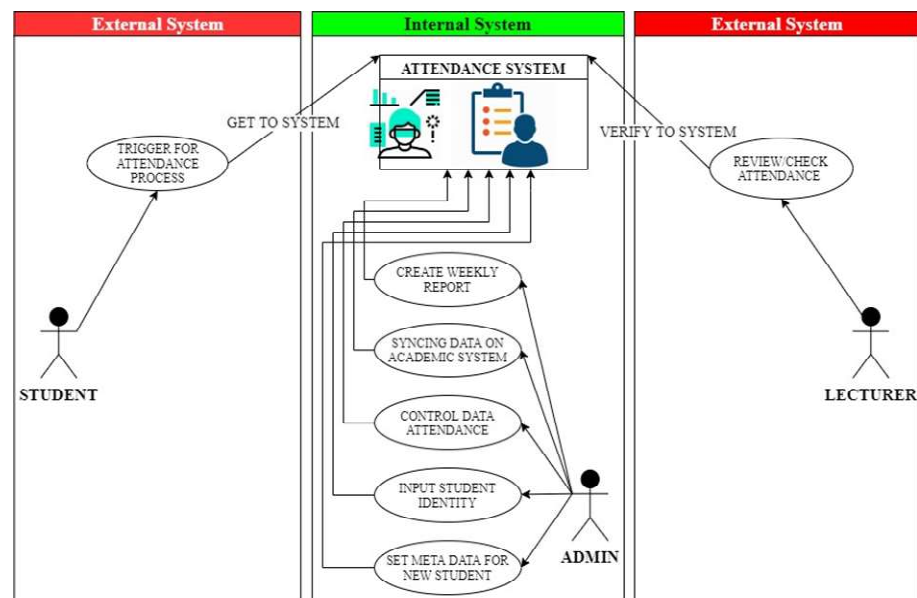


Fig. 5. Use Case Diagram of Student Absention System Using Face Recognition

III. RESULT AND DISCUSSION

Here are the test results in the attendance application system with face recognition that is already web-based, which initially uses paper.

A. Hardware Design

In this study, the attendance application system with facial recognition consists of several supporting tools which are Microprocessor Raspberry Pi 3 B, Raspberry Pi Wide Angle Fisheye Lens Camera, Memory Micro SD, Monitor, Keyboard, Mouse, Power Adapter, HDMI cable, Converter HDMI to VGA.

The hardware design consists of Raspberry Pi 3 B which becomes a microprocessor to run the processes contained in the system. Other tools that are connected with the Raspberry Pi 3 B are Memory Micro SD as a temporary data storage area, Raspberry Pi Wide Angle Fisheye Lens Camera used for taking pictures of users and as a transmission for running

face detection. Raspberry Pi 3 B is connected to the electrical source by using a Power Adapter that is specific to the Raspberry Pi 3 B with a voltage current of 24 volts.

In the Raspberry Pi 3 B the face_detector.py program has been embedded to process face detection, face_dataset.py which contains an algorithm for the dataset that training_dataset.py uses to conduct face training data. Other hardware that is connected is the monitor to display the output of the Raspberry Pi 3 B, keyboard and mouse used for facilitate control of the operation in Raspberry Pi.

Furthermore, the hardware will be connected to the internet cloud by using the wireless module in Raspberry Pi 3 B to connect to the Prima Indonesia University system which consists of a server and database. The system will be a link between hardware and the Prima Indonesia University academic system website. The Hardware Design are shown in Fig. 6.

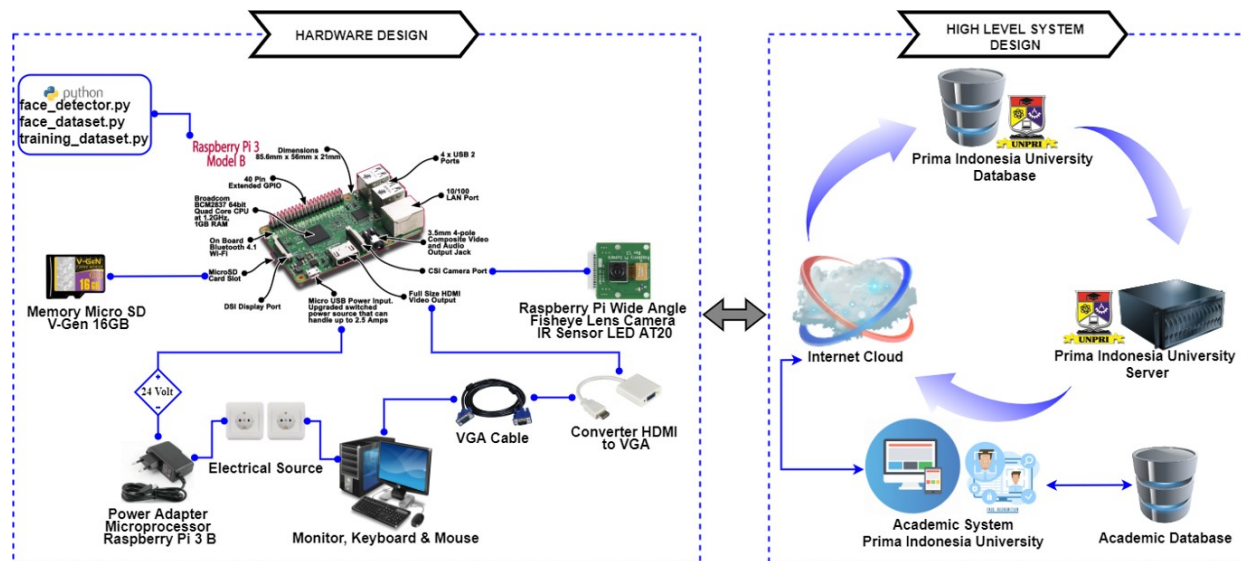


Fig. 6. Hardware Design Using Face Recognition

B. Attendance Results of Attendance Processing Tools using Raspberry Pi

To measure the effectiveness of the system that has been created, the system will be tested first by conducting a simulation of student attendance data collection. Fig. 7(a) is a display of a device that has been designed, raspberry pi has been connected to a PC to do the process of taking metadata (training data) Fig. 7(b) so that the attendance process that will run every day during class hours can already run. Fig. 7(c) is the result of taking data from students designed using the Python programming language.

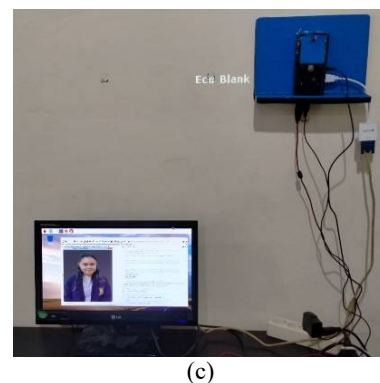
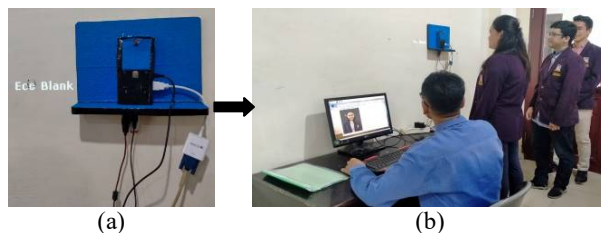


Fig. 7. Process Using Face Recognition System

C. User Interface Student Attendance Web Based

In the website section that is devoted to the process of taking attendance data from students, has several main menus. Where on the menu there is a dashboard which is the homepage of the system as shown in Fig. 8. Furthermore, there is a data input menu which is the main part of this

research. In the data input menu, there are 2 sub menus, namely the sub menu for preparing attendance and student attendance data. And what is needed for the synchronization process into the student academic system as shown in Fig. 8.

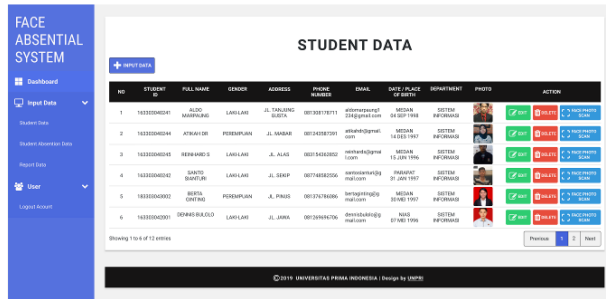


Fig. 8. GUI Student Attendance System Website

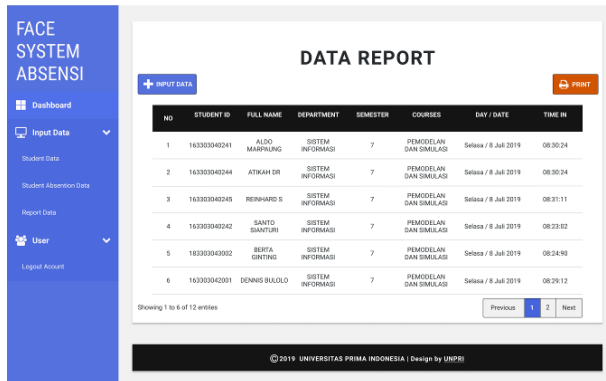


Fig. 9. GUI Student Attendance Report

To support the results of this study, an analysis of the effectiveness of the student attendance system has been carried out. The analysis process is carried out by testing the time speed of the tool to get / synchronize the faces of students during absences. Another test is a system test that is made against the time needed for the student attendance data recapitulation process. In this analysis process, the system that is built will be compared with the conditions before the student attendance system based on facial patterns built. The results for the first test regarding the timing of the tool to synchronize faces in general can be seen in the following Table I:

TABLE I. FAST TESTING OF SYNCHRONIZATION OF ABSENTION SYSTEM BASED ON FACE RECOGNITION

No.	Subject	Result	
		Manual (second)	Face Recognition System (second)
1	Student 1	00:04:25	00:02:23
2	Student 2	00:05:04	00:02:18
3	Student 3	00:04:57	00:02:54
4	Student 4	00:04:45	00:02:09
5	Student 5	00:04:36	00:02:39
6	Student 6	00:04:20	00:02:13
7	Student 7	00:04:37	00:02:14
8	Student 8	00:05:02	00:02:43
9	Student 9	00:04:52	00:02:16
10	Student 10	00:04:29	00:02:45
11	Student 11	00:04:21	00:02:36
12	Student 12	00:04:26	00:02:20
13	Student 13	00:04:45	00:02:26
14	Student 14	00:04:28	00:02:21
15	Student 15	00:05:12	00:02:29

16	Student 16	00:04:50	00:02:52
17	Student 17	00:04:09	00:02:19
18	Student 18	00:04:54	00:02:17
19	Student 19	00:04:41	00:02:40
20	Student 20	00:04:23	00:02:36
Average Value		00:04:40	00:02:28

In Table I, it can be seen that the simulation testing of the system was carried out using 20 students as the test subjects. Each student will attend attendance using an attendance system based on facial recognition patterns and attendance manually with attendance sheets. The average time needed for the system to synchronize students is 00:02:28 seconds. The time parameter is implicitly the time of attendance in the system. Whereas for the manual use of the signature process on a sheet of paper attendance has an average time of 00:04:40 seconds. Sometimes the implementation of a sheet of paper can be held in one of the students (not executed) so that it will hamper the attendance process. Based on these data it is certain that the use of attendance systems based on facial pattern recognition is more effective and efficient in time and effort.

The second test is a test of recapitulation of student attendance lists using attendance systems based on face patterns and manually. The test is as shown in table II.

TABLE II. TESTING RECAPITULATION OF STUDENT ATTENDANCE USING A SYSTEM BASED ON FACE RECOGNITION

No.	Weeks	Time Needed to Recapitulation Students Attendance	
		Manual (Days)	Face Recognition System (Days)
1	I	6	4
2	II	7	4
3	III	6	3
4	IV	8	4

In Table II it can be seen that the simulation test of the system is carried out within a month with the target of recapitulating student attendance within a week. Attendance system based on facial patterns that are built can provide a faster time attendance data recapitulation of students per week. With a range manually it takes 6-8 days and if using the system it only takes 3-4 days. This can improve the performance of the existing academic system.

IV. CONCLUSION

The student attendance system based on facial patterns has been successfully built and implemented. The resulting system has fulfilled the initial purpose of the study which aims to create an effective system in recording and recording attendance lectures at the University of Prima Indonesia. This is in accordance with the previous analysis process where the time needed to process the student attendance data recap before the system is made in 6-8 days, and after the system is built into 3-4 days.

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